

Q and P: Institutions, Agents, and Stopping Rules

Q Competitive Market Behavior of LLMs [arXiv:2609.02580](#)

P Adversarial Review: Structured Disagreement for Grounded Agentic Code Review [arXiv:2608.18167](#)

Research phase: PAUSE-ON-ITERATE after 3 rounds, 3 ITERATE, 0 REVISE

The pair in two sentences

Q studies LLM traders in a continuous double auction and reports slower or absent convergence and lower allocative efficiency than in the human benchmark. P studies Adversarial Review (AR), where a reviewer and critic exchange structured judgments before a coding agent revises its work, and reports gains alongside a false-consensus failure mode.

The connexion pursued

The pursued question was whether observed multi-agent performance comes from agent information or from the institution, especially its stopping rule and admissible messages. A zero-intelligence control can separate those contributions: Q permits a constrained random-trader benchmark, while P suggests analogous controls in which critic verdicts are random or the reviewer alone is allowed to iterate (ledger entries 11, 14, 16, and 18).

What was found

The firm result is on Q alone. Two independent implementations of Q’s stated mechanism give constrained zero-intelligence traders (ZI-C) efficiency near 0.96, about 5.8–5.9 trades per round, and mean transaction price 2.00; every reported LLM cell in Q has efficiency 0.36–0.91 (ledger entries 3 and 5). The comparison is robust to the simulated quote range (ledger entry 5). On a finer horizon grid, the five-round LLM means equal the ZI-C efficiency obtained after only about 53 draws for GPT Large, 80 for Gemini Large, and 131 for GPT Small, although Q allows 300 draws (ledger entry 8).

The surplus accounting explains the benchmark. With maximum feasible surplus 7.50, define

$$1 - E = O + C,$$

where O is the foregone surplus of intramarginal traders left out, divided by 7.50, and C is the surplus destroyed by extramarginal participation, divided by 7.50. The five efficient pairs contribute 2.50, 2.00, 1.50, 1.00, 0.50, so at most $k = 1, \dots, 5$ completed efficient pairs yield efficiencies 0.33, 0.60, 0.80, 0.93, 1.00 (ledger entries 4 and 7). ZI-C has $O = 0.009$ and $C = 0.032$, whereas a stylised one-cent “creeper” has omission about 0.82; making acceptance the default raises volume but produces commission 0.10–0.17 and efficiency only 0.71–0.79 (ledger entry 4). Thus a cheap terminal move can exchange missed trades for bad trades without recovering the ZI-C result. The direction is model-dependent: reported GPT Large and Gemini Large mainly omit trades, but GPT Small and unreported Gemini runs also display commission (ledger entries 6 and 7).

This does not establish the proposed connexion to P. In P, Single-reviewer and Two-reviewers edit once, while MARS and AR re-review revised artifacts; those latter methods are also the methods above the 75–77% cluster (ledger entries 11, 16, and 22). Consequently P does not identify the informational value of reviewer–critic interaction separately from re-review, stopping, and adaptive compute. A four-parameter toy model shows only plausibility: when one review–edit cycle is calibrated to have zero net effect, iteration until approval adds 0.3–9.1 percentage points, with median 2.6, but it is not evidence about P’s tasks (ledger entry 12).

Objections and resolutions

The first objection was that an enforced tick-size result had been attributed too directly to Q. Code inspection showed that Q uses strict floating-point improvement for LLM orders and applies its price increment only in a ZI deadlock check; the tick experiment therefore concerns a new enforced and stated message constraint, not Q as run (ledger entries 9 and 13). This objection was resolved by narrowing the claim.

The second objection was that ZI-C and the LLMs might not share the same effective constraint. Q's exchange does not enforce reservation values, although its prompt tells LLMs never to violate them; violation events are logged but their rate is not reported (ledger entries 13, 18, 20, and 22). The 0.96 result is therefore a constrained-strategy floor conditional on LLM prompt compliance, not an unconditional exchange baseline.

The verifier objected that the Q-P connexion remained analogy rather than a supported cross-paper result and requested a matched P ablation (ledger entry 24). Both peers accepted the objection. They also established that the supplied material lacks P's task instances, per-task outcomes, traces, disagreement rates, orchestration, and credentials, so running that ablation here would invent evidence (ledger entries 25, 27, and 29). P also reports MARS as 82% in Table 1 but calls it 85% in prose, an unresolved internal inconsistency (ledger entries 16 and 22).

Sources outside Q and P

The work used Gode and Sunder's 1993 ZI-C strategy as background and inspected Q's released repository, including its configurations, notebook, and order-handling code (ledger entries 4, 6, and 13). A prior-work search found BAZAAR, a sealed-bid double auction that reports individual profit rather than market efficiency; it is a different comparison and supplies no novelty claim here (ledger entry 14). No outside source supplied evidence for the proposed P ablation.

What remains open

The material is thin on the cross-paper question. No LLM intervention was run; Q's LLM reservation-violation rate is unknown; two author ZI-C batches have unexplained price-dispersion differences; and P provides neither uncertainty nor paired task-level outputs (ledger entries 14, 18, 27, and 31). The supported conclusion is therefore negative: Q supplies a standalone constrained ZI-C result, but Q and P together do not yet establish a cross-paper advance (ledger entries 27, 29, and 31).

Smallest settling step

On the same P tasks, compare an R-only iterate-to-clean loop, AR, and AR with a critic whose verdict is random but matched to the observed disagreement rate. Hold outer caps and compute equal and report paired per-task outcomes. If AR beats both controls, critic information has an effect beyond re-review and stopping; otherwise the proposed institutional decomposition is not supported (ledger entries 25, 27, and 31).